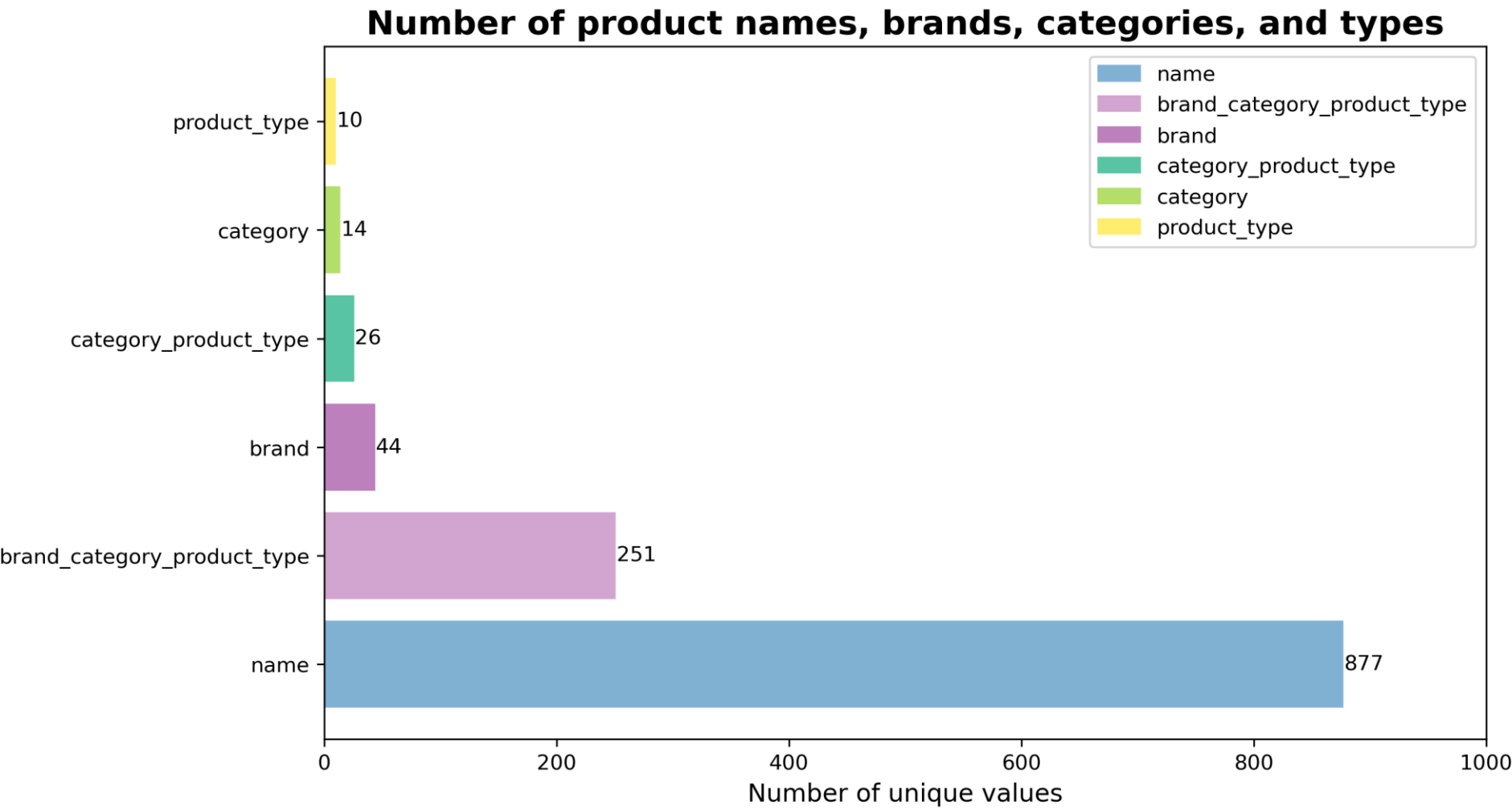
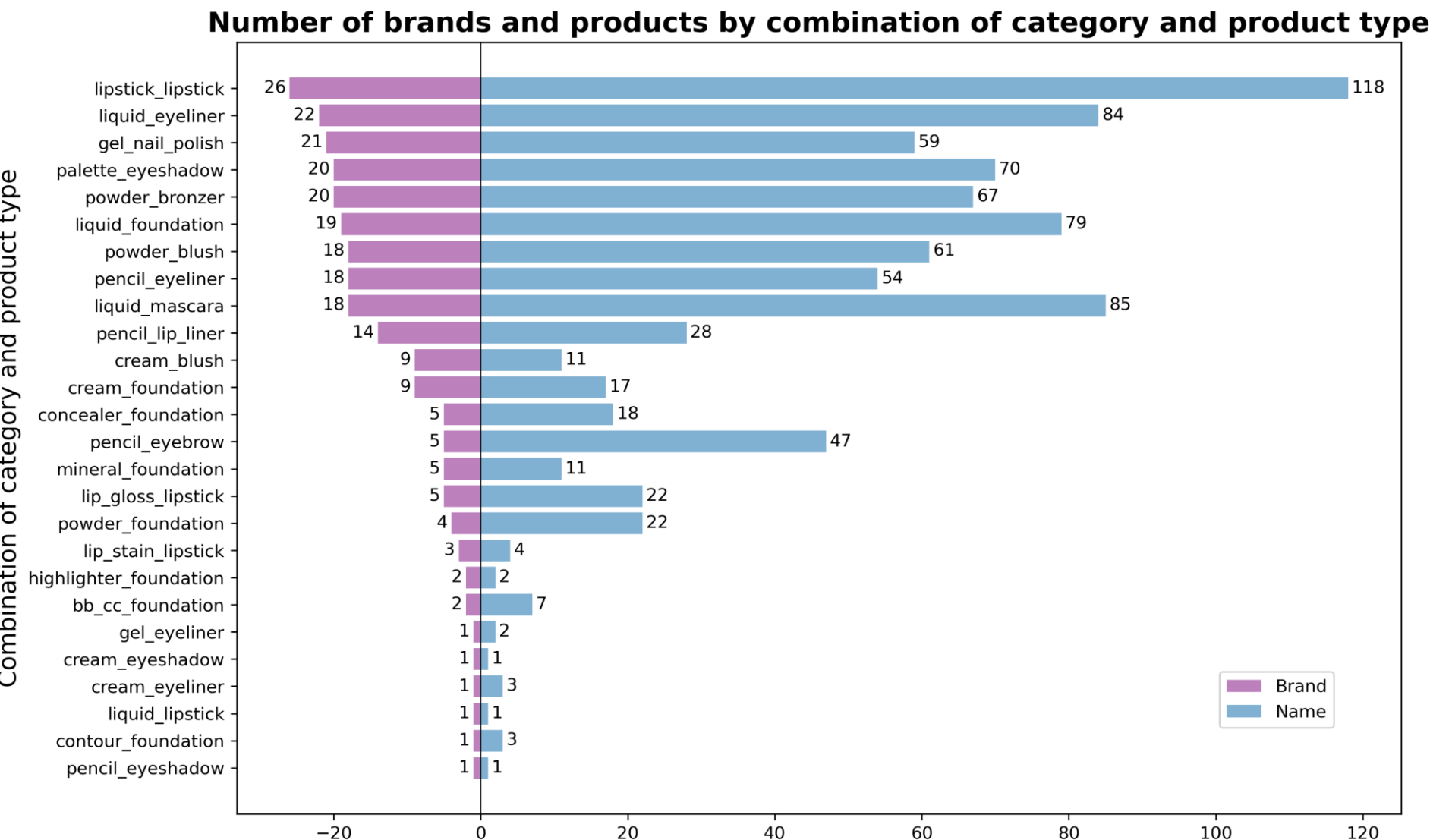


Number of Unique Products, Brands, Categories, and Types



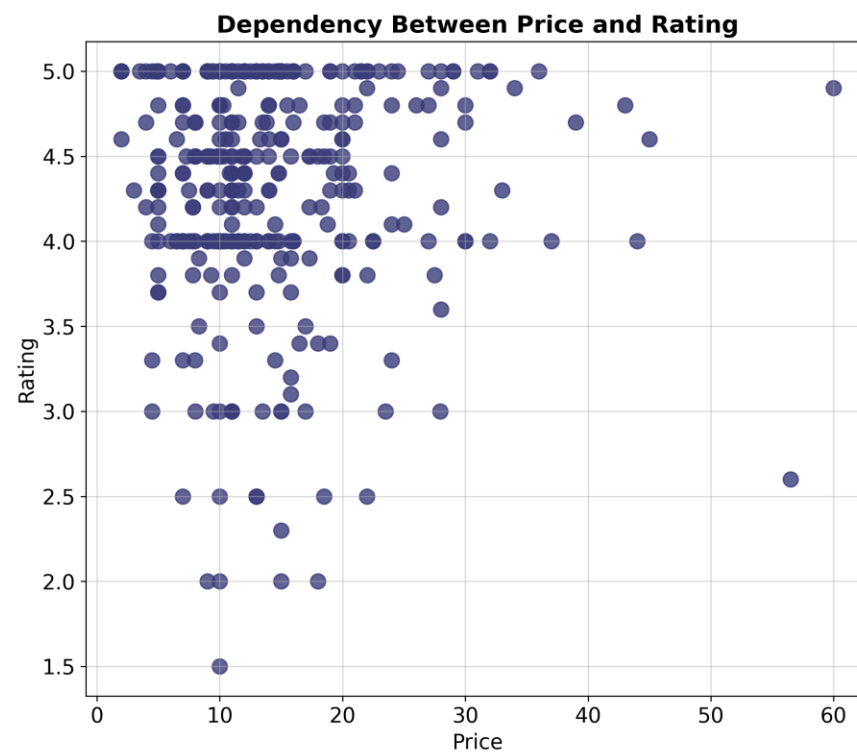
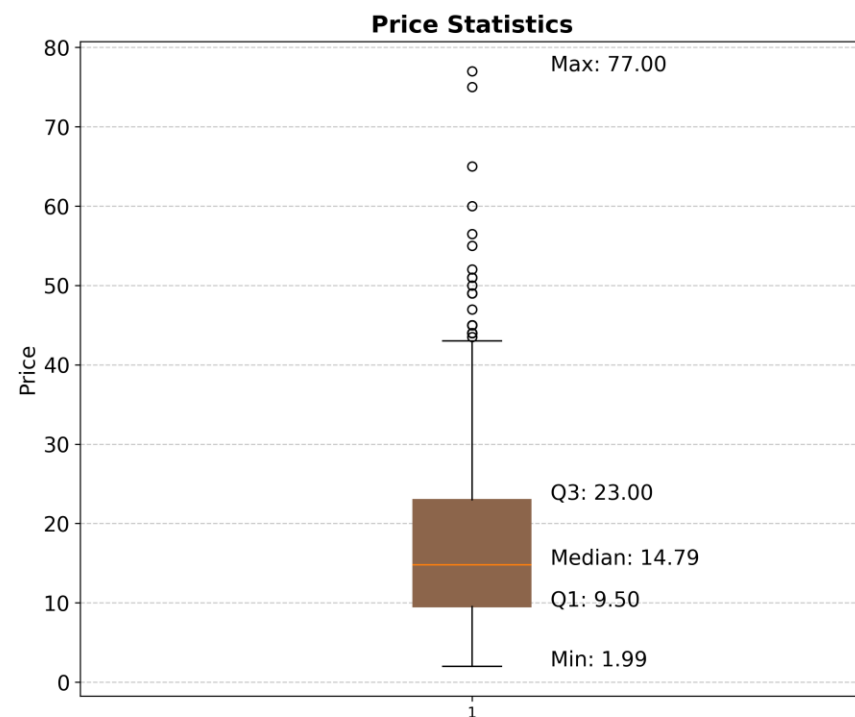
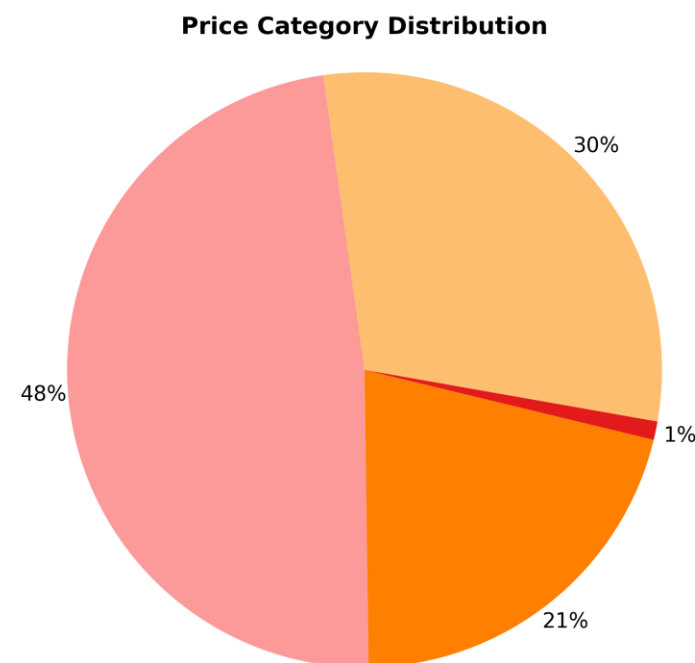
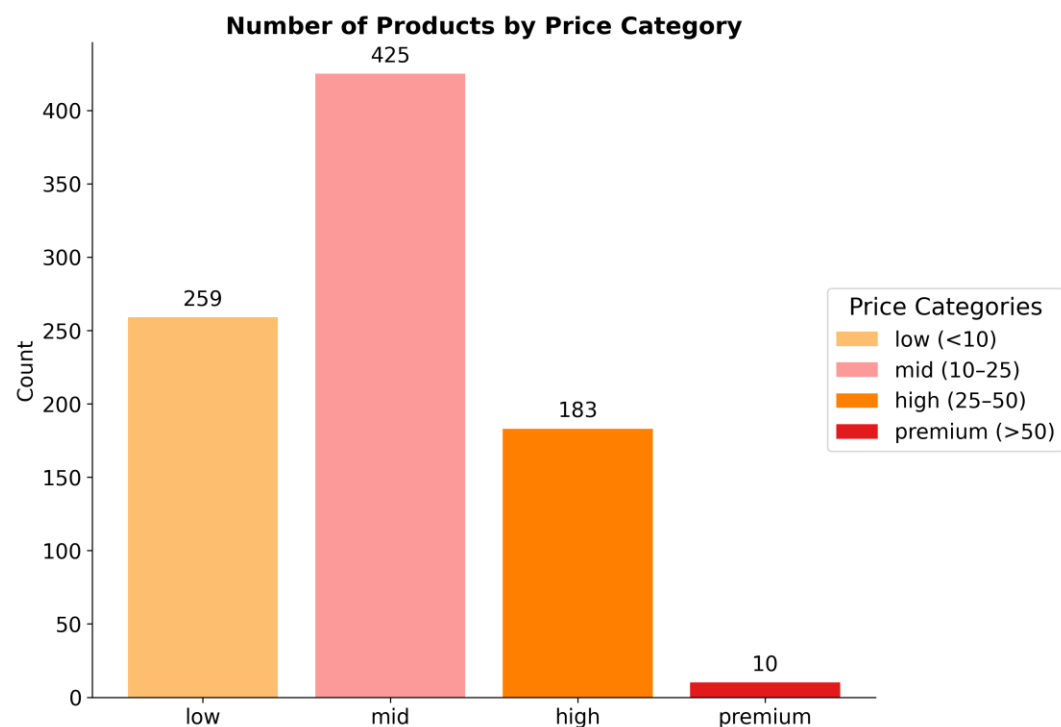
- Chart type: barh (horizontal bar chart).
- Displayed: Number of unique values for each data category.

Number of Brands and Products by Categories



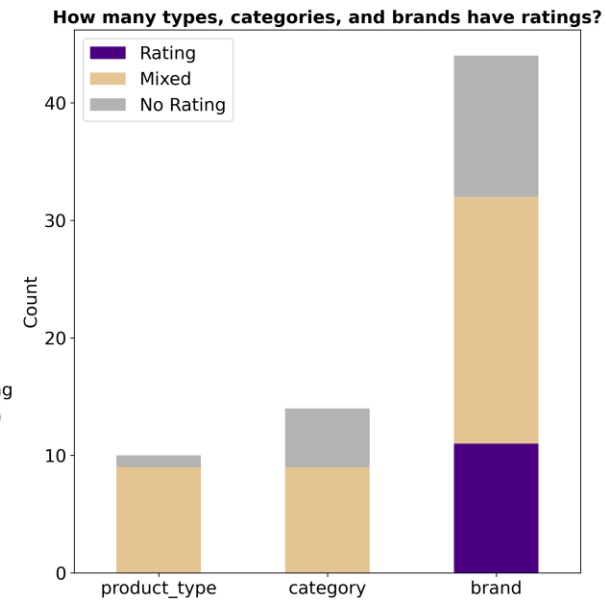
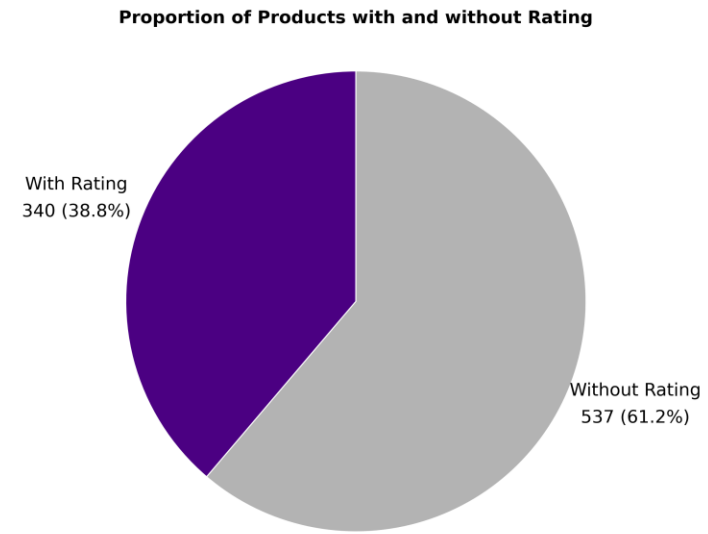
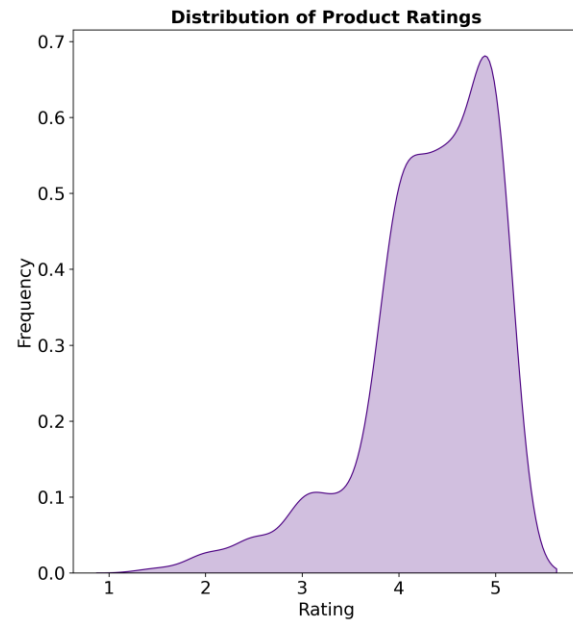
- Chart type: barh (horizontal bars).
- Displayed:
 - Brand — Number of unique brands (on the left).
 - Name — Number of unique products (on the right) for each combination of category and product type.

Price Analysis



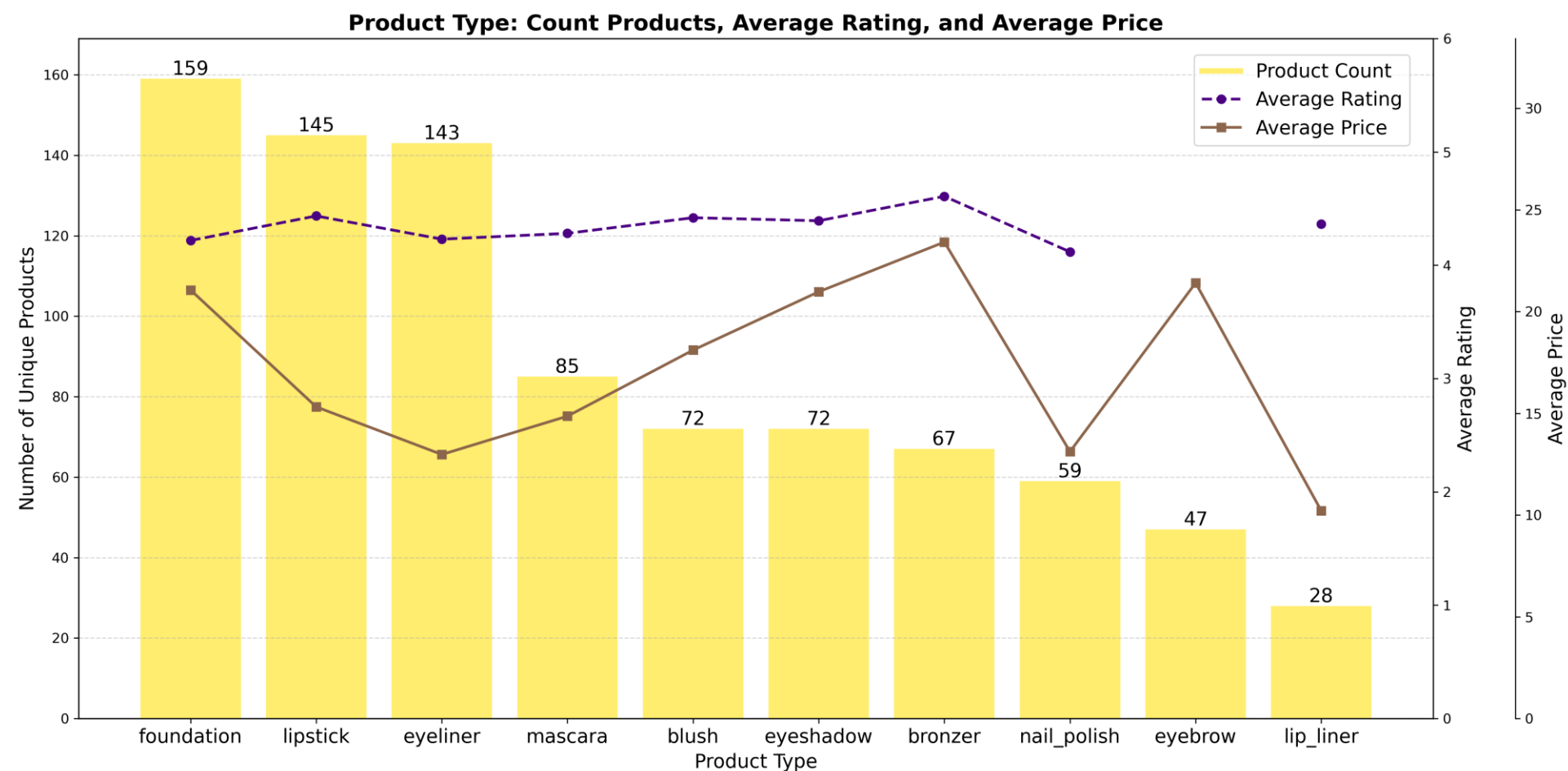
- Chart types: bar, pie, boxplot, scatter.
- Displayed:
 - Bar chart: Number of items by price categories.
 - Pie chart: Percentage share of each price category.
 - Boxplot: Minimum, maximum, median price, quartiles.
 - Scatter plot: Dependency between price and rating.

Product Ratings



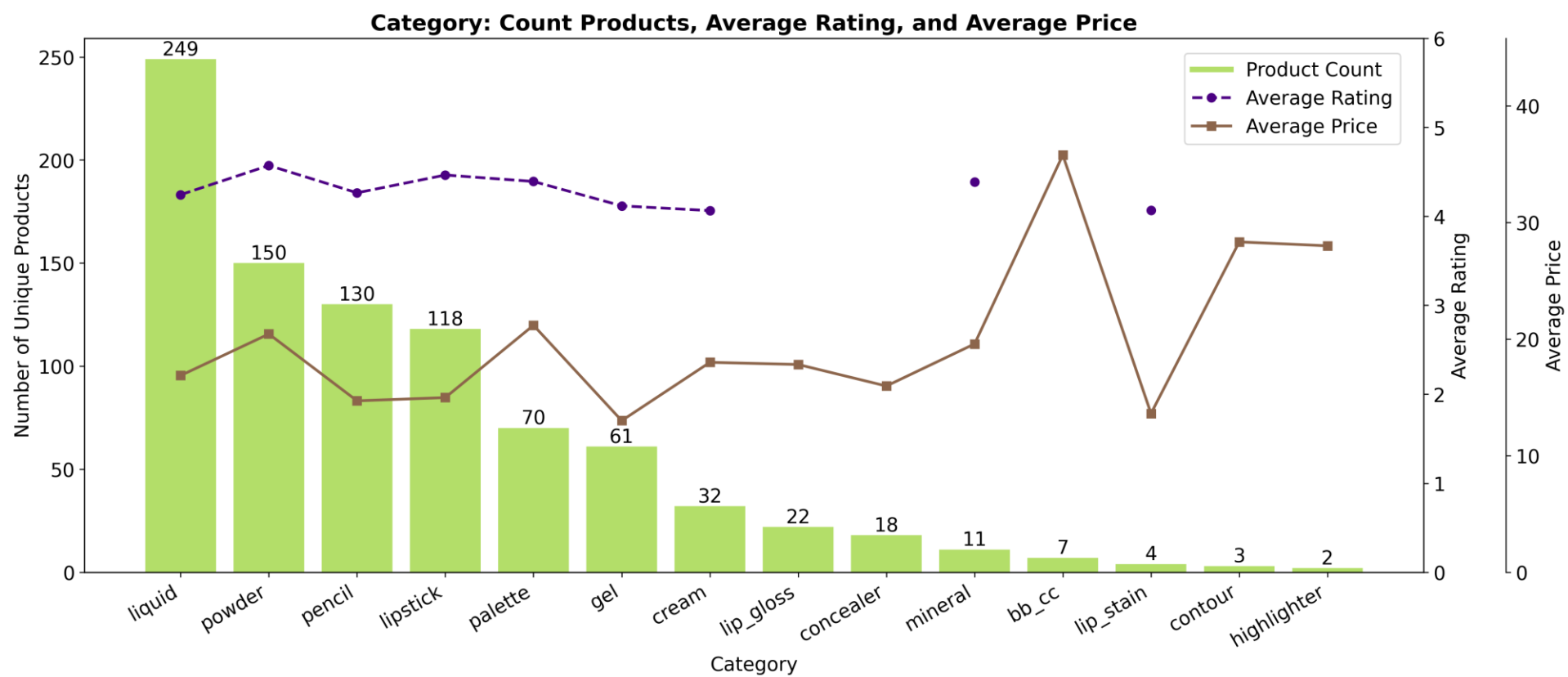
- Chart types: kdeplot (density estimation), pie, bar.
- Displayed:
 - Distribution of rating values.
 - Share of items with and without ratings.
 - Count of types, categories, and brands with ratings.

Product Type Analysis



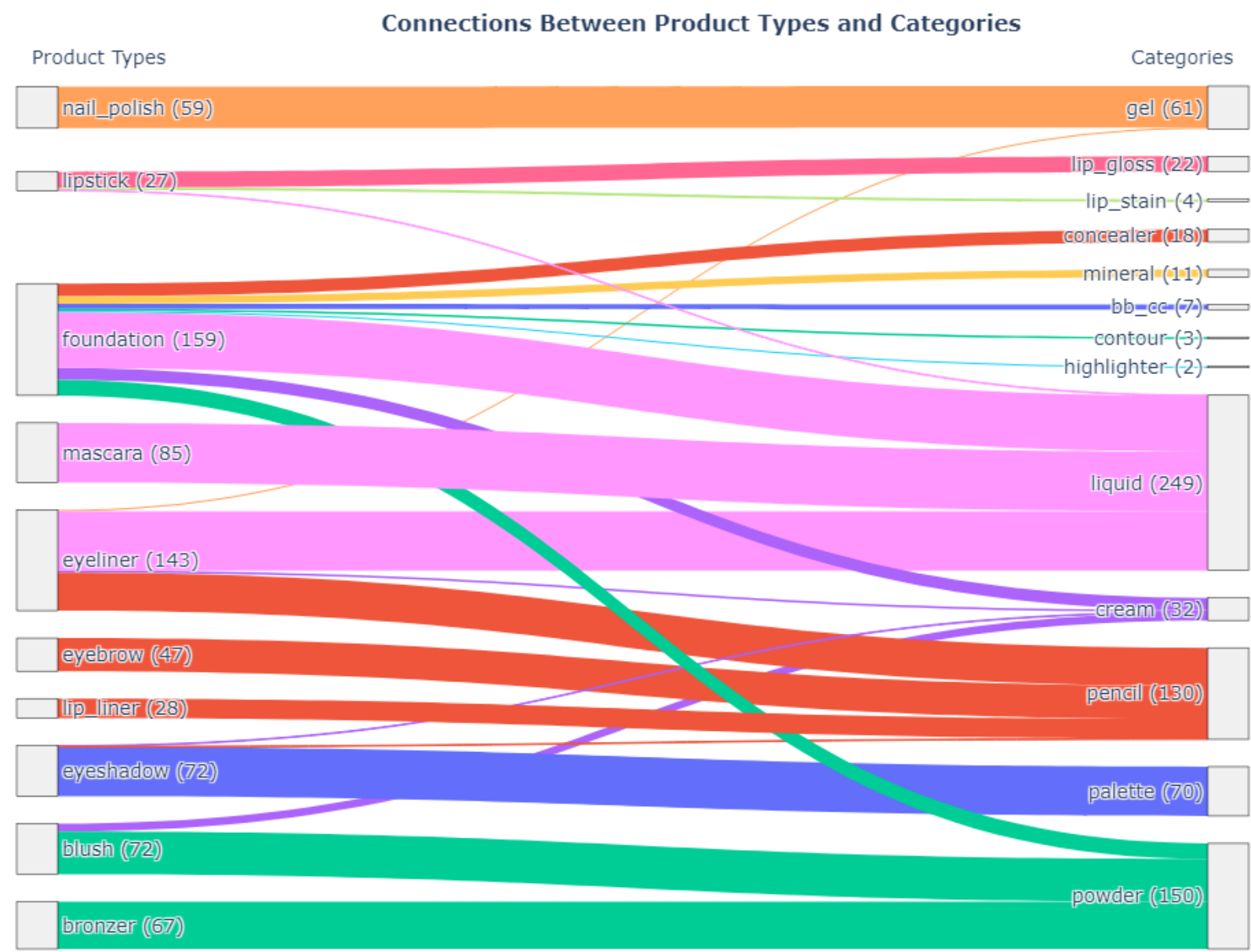
- Chart types: bar (bars), line (lines).
- Displayed:
 - Number of products for each type.
 - Average rating and price by product types.

Product Categories



- Chart types: bar (bars), line (lines).
- Displayed:
 - Number of products.
 - Average rating and price by categories.

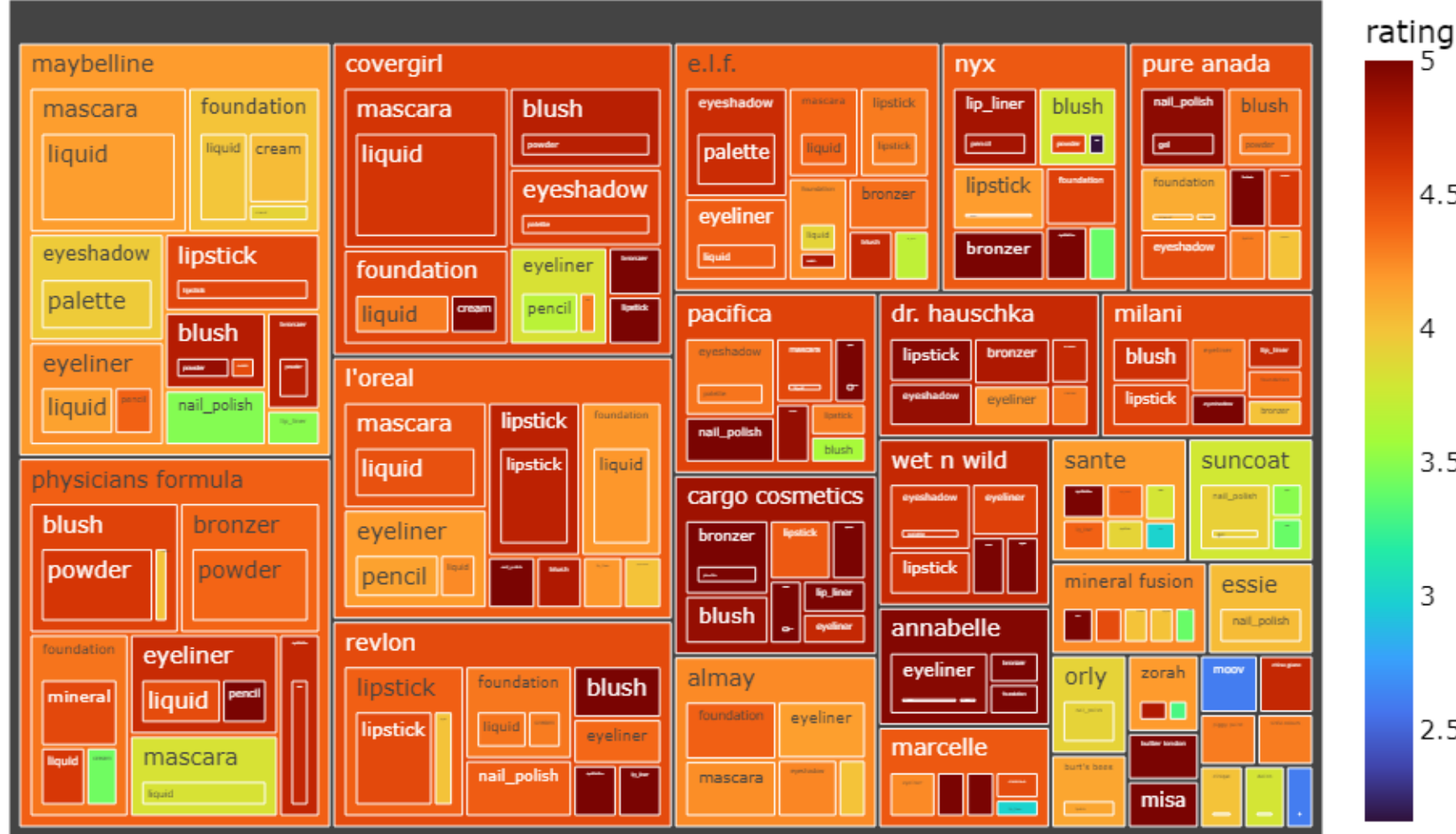
Connections Between Product Types and Categories



- Chart type: Sankey Diagram.
- Displayed:
 - Relationships between product types and categories.
 - Distribution of products between types and categories represented by line widths.

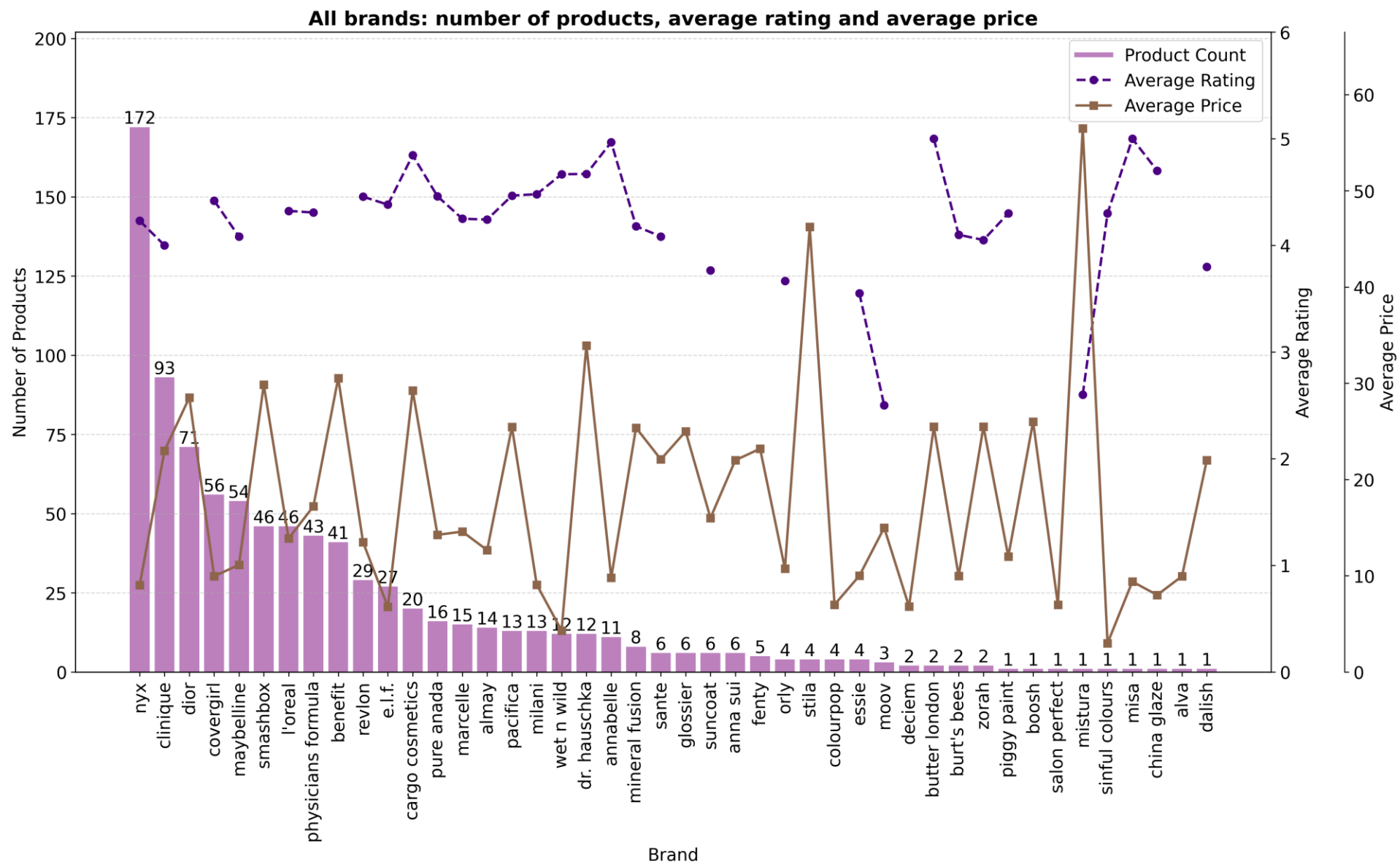
Brands by Categories and Product Types

Brands by Categories and Product Types with Ratings



- Chart type: Treemap.
- Displayed:
 - Hierarchical structure of brands in categories.
 - Rating indicated by a color scale.

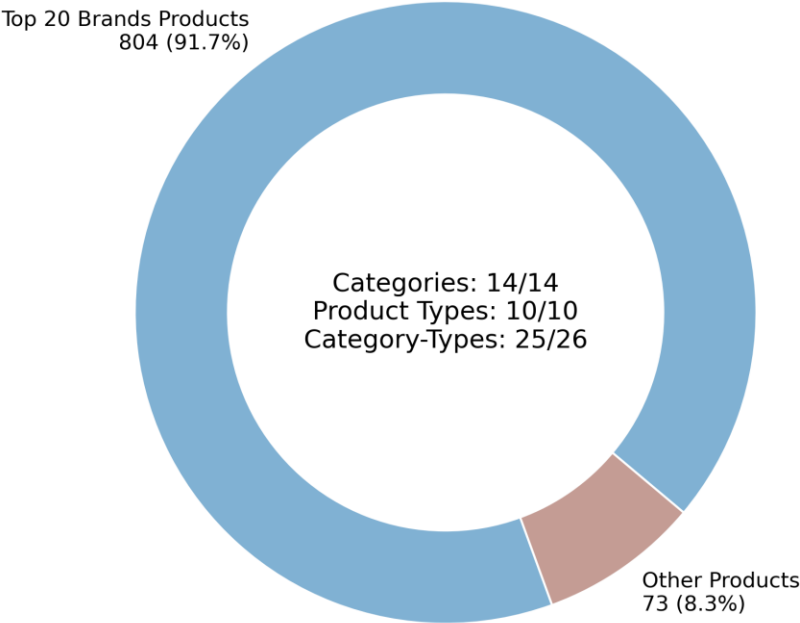
Overview of All Brands



- Chart types: bar, line.
- Displayed:
 - Number of products for each brand.
 - Average rating and price.

Top 20 Brands

Coverage of Top 20 Brands in Product Distribution



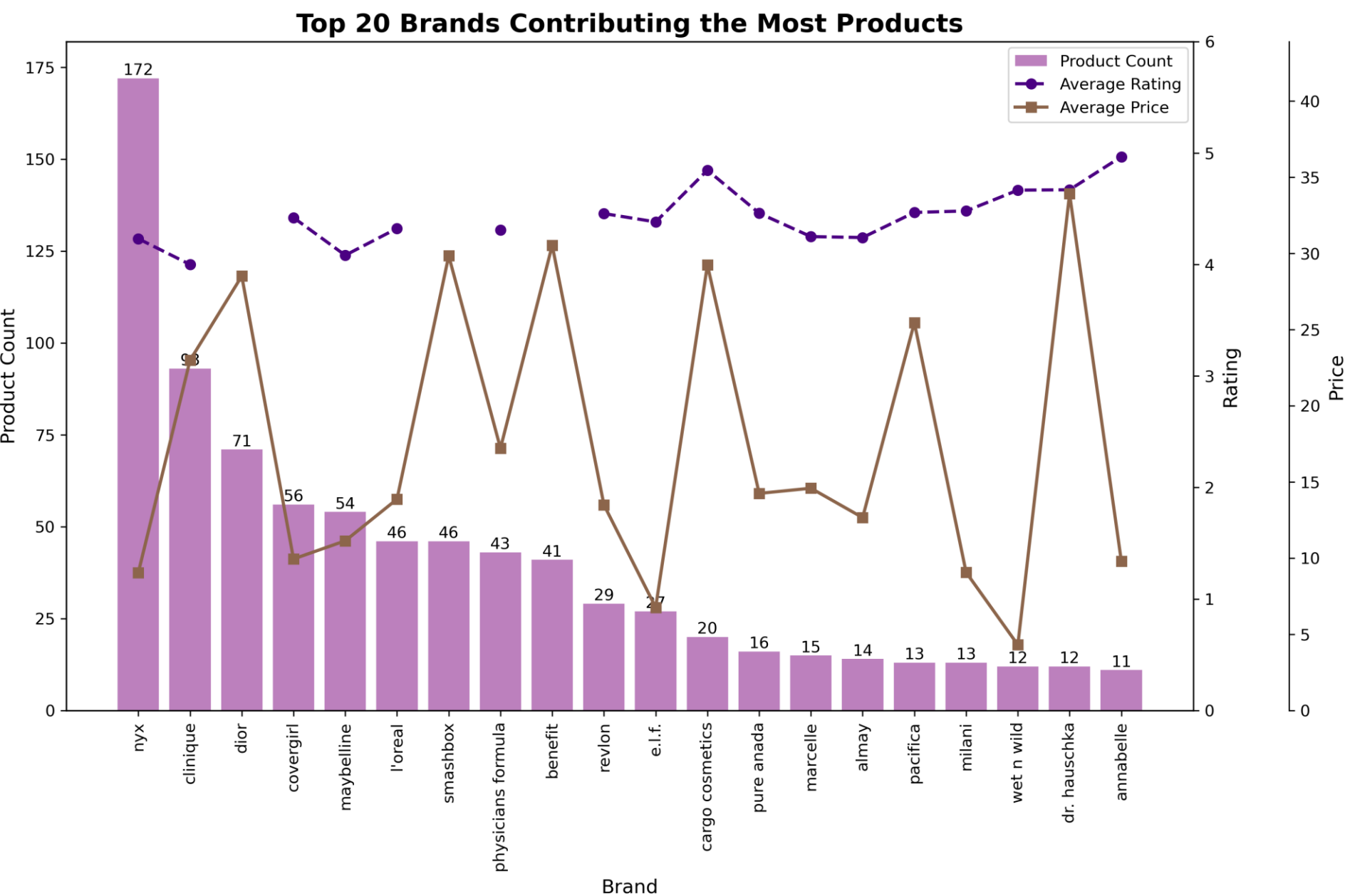
Visualization of Excluded Product

Displays Category_product_type and the products it contains, which are not included in the top 20 brands' data.



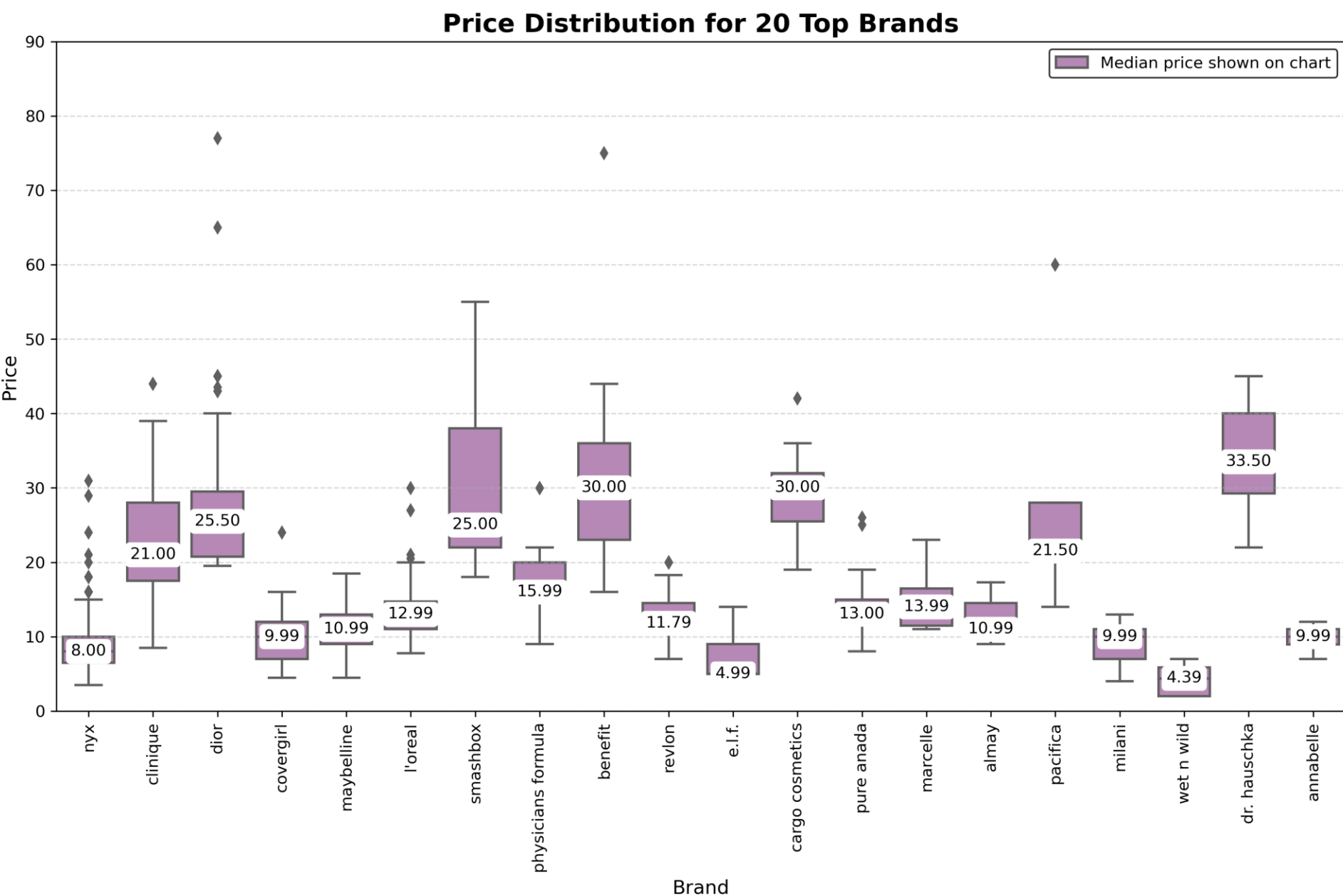
- Chart types: donut chart, pie chart.
- Displayed:
 - Share of products offered by the top 20 brands.
 - Count of categories, product types, and their coverage.

Analysis of Top 20 Brands



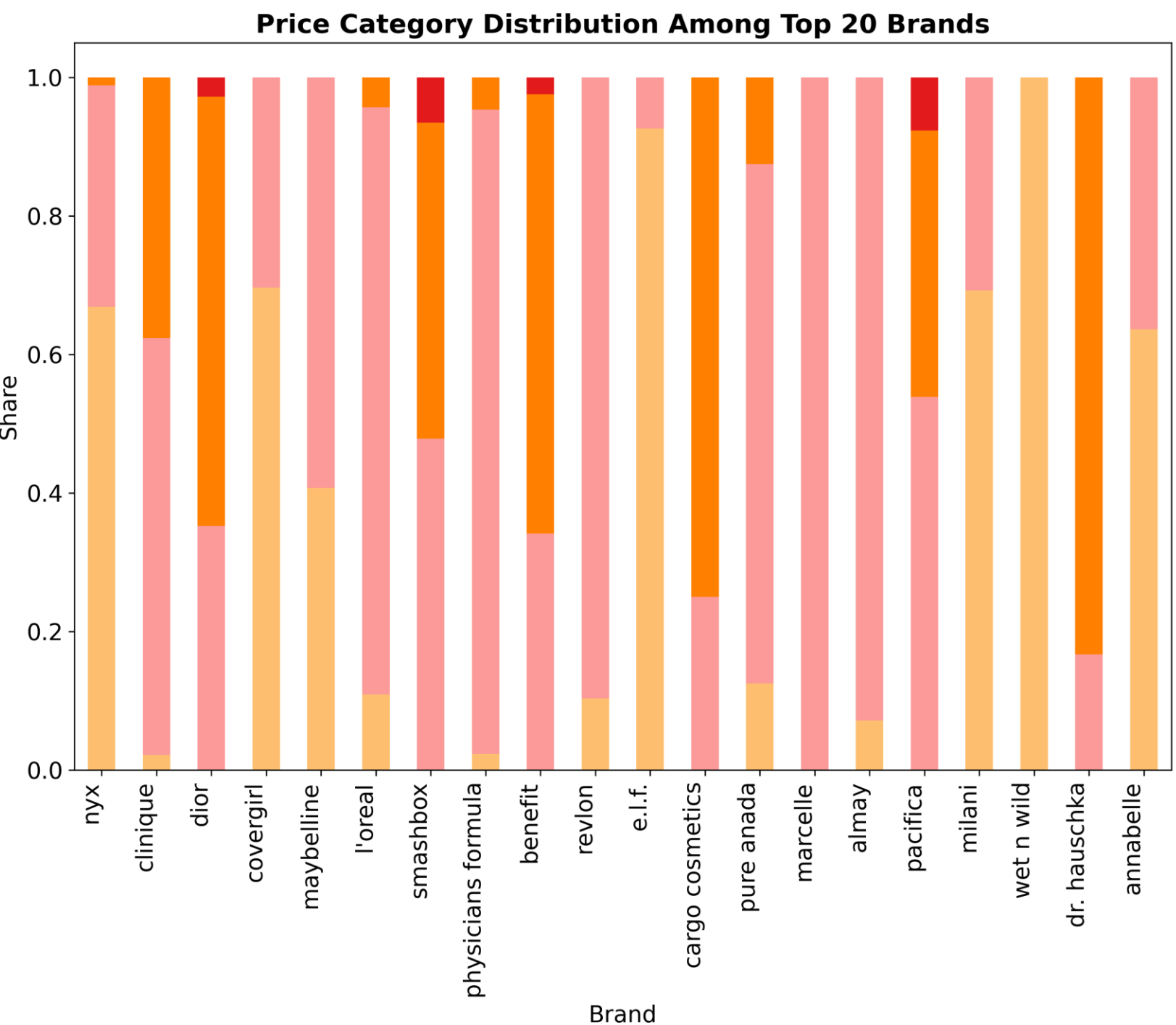
- Chart types: bar, line.
- Displayed:
 - Number of products, ratings, and average price per brand.

Top Brands Prices



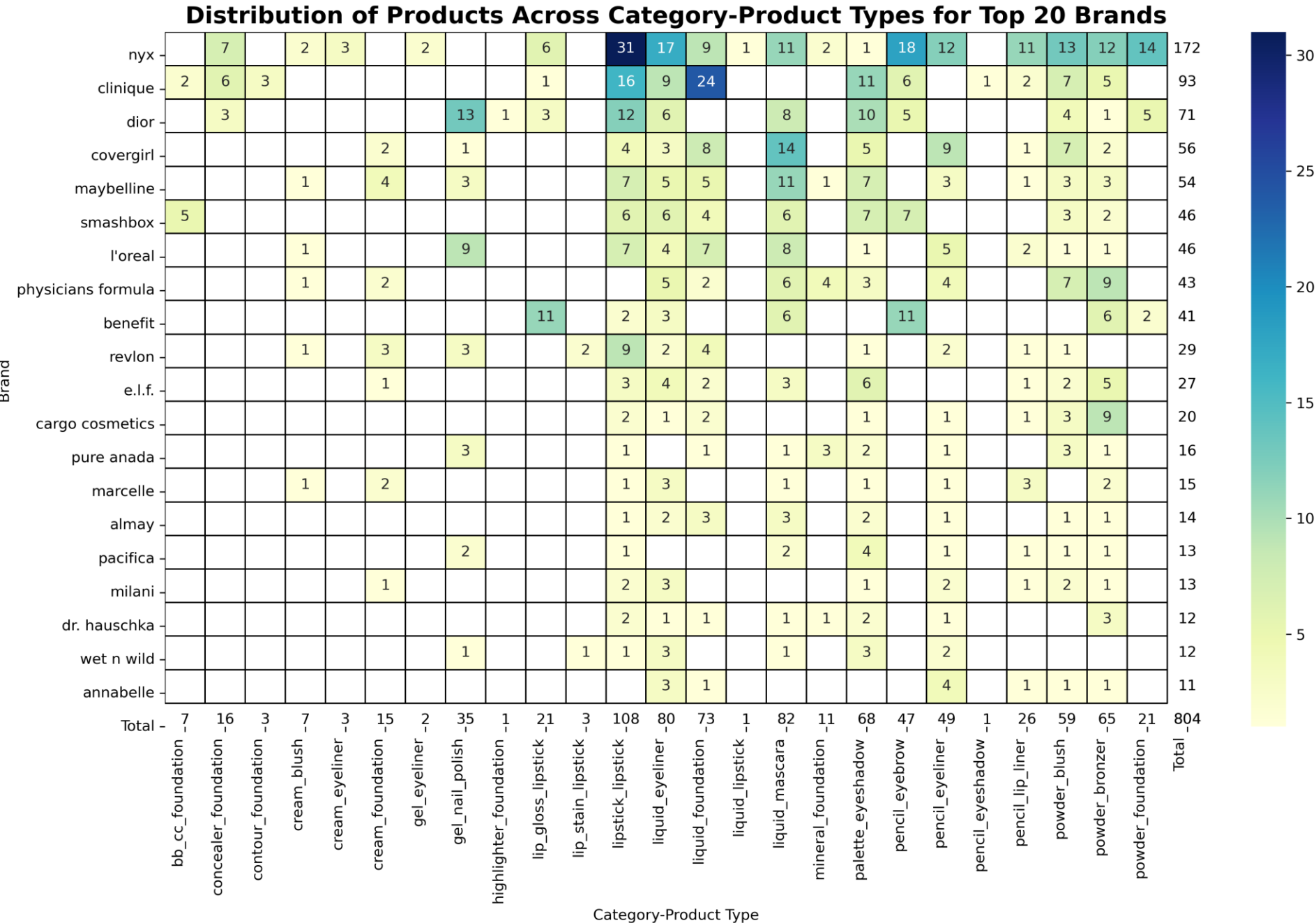
- Chart type: boxplot.
- Displayed:
 - Price distribution for each brand, including minimum, maximum, median.

Price Distribution Among Products



- Chart type: stacked bar.
- Displayed:
 - Representation of brands across different price categories.

Product Distribution Across Categories and Types Among Top 20 Brands



- Chart type: heatmap.
- Displayed:
 - Count distribution of category-product types among top 20 brands.
 - The chart highlights which brands actively operate across different category-product types and their relationships.